



Certified Kubernetes Administrator (CKA)

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Section 21 - Lightning Labs

- *Short, hands-on exercises designed to rapidly build and reinforce Kubernetes skills.*

Intro

Welcome to the KodeKloud CKA Lightning Labs!

The screenshot displays the KodeKloud CKA Lightning Labs interface. On the left, a 'Task' panel shows a progress bar at 0% and a score of 80%. Below this, three tasks (Q.1, Q.2, Q.3) are listed, each with 'Question' and 'Solution' tabs. Q.1 and Q.3 are marked 'Incomplete', while Q.2 is marked 'Task completed?'. The 'Solution' tab for Q.2 shows a terminal snippet with the command: `ace --sort-by=.metadata.name > /opt/admin2406_data`. On the right, a 'Terminal 1' window is open, displaying a welcome message and a terminal prompt: `controlplane ~ →`. A 'Feedback' button is visible on the far right.

KodeKloud

Task

05:27

0%

Your score
Pass Percentage - 80%

Q.1

Question

Incomplete ✕

Solution

Details

Q.2

Question

Incomplete ✕

Solution

Run the below command to get the correct output:

```
ace --sort-by=.metadata.name > /opt/admin2406_data
```

Details

✕ Task completed?

Q.3

Question

Incomplete ✕

Solution

Details

Terminal 1

Welcome to the KodeKloud Hands-On lab

All rights reserved

controlplane ~ →

Feedback

Lightning Lab

Hands-on practice to quickly apply and reinforce key Kubernetes concepts.

Q&A - 1

1 2 3 4 5 6 7



Weight: 15

Upgrade the current version of kubernetes from 1.26.0 to 1.27.0 exactly using the kubectl utility. Make sure that the upgrade is carried out one node at a time starting with the controlplane node. To minimize downtime, the deployment gold-nginx should be rescheduled on an alternate node before upgrading each node.

Upgrade controlplane node first and drain node node01 before upgrading it. Pods for gold-nginx should run on the controlplane node subsequently.

1
● Cluster Upgraded?

2
● pods 'gold-nginx' running on controlplane?

```
controlplane ~ ❌ k get pod -A -o wide | grep gold-nginx
default      gold-nginx-6c5b9dd56c-nj8xw    1/1    Running    0          9m58s    10.244.192.1    node01    <none>    <none>

controlplane ~ ➔ k describe node controlplane | grep Taint
Taints:      node-role.kubernetes.io/control-plane:NoSchedule

controlplane ~ ➔ k describe deployment gold-nginx | grep -i tolerations

controlplane ~ ❌
```

1. Drain node là phải đứng ở controlplane

2. Đọc kĩ đề, phải remove cái Taint ở controlplane (Nếu ko remove từ đầu thì khi double check tiến hành remove và drain node01 cũng đc)

Q&A - 1

```
controlplane ~ ❌ k get nodes
```

NAME	STATUS	ROLES	AGE	VERSION
controlplane	Ready	control-plane	136m	v1.27.0
node01	Ready	<none>	135m	v1.26.0

```
controlplane ~ ➔ kubectl drain node01 --ignore-daemonsets
```

```
node/node01 cordoned
```

```
Warning: ignoring DaemonSet-managed Pods: kube-system/kube-proxy-hphm5, kube-system/weave-net-xxn67
```

```
evicting pod kube-system/coredns-787d4945fb-cgcll
```

```
evicting pod admin2406/deploy4-55554b4b4c-886kp
```

```
evicting pod default/gold-nginx-6c5b9dd56c-bg554
```

```
evicting pod admin2406/deploy2-7b6d9445df-k5nz5
```

```
evicting pod admin2406/deploy3-66785bc8f5-66q7n
```

```
evicting pod admin2406/deploy1-5d88679d77-cbbsg
```

```
controlplane ~ ❌ k drain node01 --ignore-daemonsets
```

```
node/node01 already cordoned
```

```
Warning: ignoring DaemonSet-managed Pods: kube-system/kube-proxy-hphm5, kube-system/weave-net-xxn67
```

```
evicting pod default/gold-nginx-6c5b9dd56c-nj8xw
```

```
evicting pod admin2406/deploy2-7b6d9445df-2slhb
```

```
evicting pod admin2406/deploy5-7cbf794564-tp8ts
```

```
evicting pod admin2406/deploy4-55554b4b4c-dc7sf
```

```
evicting pod admin2406/deploy3-66785bc8f5-f9bl6
```

```
evicting pod admin2406/deploy1-5d88679d77-ll5wj
```

```
pod/gold-nginx-6c5b9dd56c-nj8xw evicted
```

```
pod/deploy3-66785bc8f5-f9bl6 evicted
```

```
pod/deploy4-55554b4b4c-dc7sf evicted
```

```
pod/deploy2-7b6d9445df-2slhb evicted
```

```
pod/deploy5-7cbf794564-tp8ts evicted
```

```
pod/deploy1-5d88679d77-ll5wj evicted
```

```
node/node01 drained
```

```
controlplane ~ ➔ k get pod -A -o wide | grep gold-nginx
```

default	gold-nginx-6c5b9dd56c-v9sn7	0/1	ContainerCreating	0	12s	<none>	controlplane	<none>	<none>
---------	-----------------------------	-----	-------------------	---	-----	--------	--------------	--------	--------

```
controlplane ~ ➔ k get pod -A -o wide | grep gold-nginx
```

default	gold-nginx-6c5b9dd56c-v9sn7	0/1	ContainerCreating	0	17s	<none>	controlplane	<none>	<none>
---------	-----------------------------	-----	-------------------	---	-----	--------	--------------	--------	--------

```
controlplane ~ ➔ k get pod -A -o wide | grep gold-nginx
```

default	gold-nginx-6c5b9dd56c-v9sn7	1/1	Running	0	20s	10.244.0.9	controlplane	<none>	<none>
---------	-----------------------------	-----	---------	---	-----	------------	--------------	--------	--------

```
controlplane ~ ➔ k uncordon node01
```

```
node/node01 uncordoned
```

```
controlplane ~ ➔ k get nodes
```

NAME	STATUS	ROLES	AGE	VERSION
controlplane	Ready	control-plane	157m	v1.27.0
node01	Ready	<none>	157m	v1.27.0

15%

Your score
Pass Percentage - 80%

Q. 1

 Question

Complete 

Upgrade the current version of kubernetes from **1.26.0** to **1.27.0** exactly using the **kubeadm** utility. Make sure that the upgrade is carried out one node at a time starting with the controlplane node. To minimize downtime, the deployment **gold-nginx** should be rescheduled on an alternate node before upgrading each node.

Upgrade **controlplane** node first and drain node **node01** before upgrading it. Pods for **gold-nginx** should run on the **controlplane** node subsequently.

Q&A - 2



1 2 3 4 5 6 7

Weight: 15

Print the names of all deployments in the `admin2406` namespace in the following format:

```
DEPLOYMENT CONTAINER_IMAGE READY_REPLICAS NAMESPACE
```

```
<deployment name> <container image used> <ready replica count> <Namespace>
```

. The data should be sorted by the increasing order of the `deployment name` .

Example:

```
DEPLOYMENT CONTAINER_IMAGE READY_REPLICAS NAMESPACE
```

```
deploy0 nginx:alpine 1 admin2406
```

Write the result to the file `/opt/admin2406_data` .

● Task completed?

Q&A - 2

```
controlplane ~ → k get deployments -n admin2406 --sort-by=.metadata.name -o custom-columns=DEPLOYMENT:.metadata.name,CONTAINER_IMAGE:.spec.template.spec.containers[*].image,READY_REPLICAS:.status.readyReplicas,NAMESPACE:.metadata.namespace > /opt/admin2406_data
```

```
controlplane ~ → cat /opt/admin2406_data
```

DEPLOYMENT	CONTAINER_IMAGE	READY_REPLICAS	NAMESPACE
deploy1	nginx	1	admin2406
deploy2	nginx:alpine	1	admin2406
deploy3	nginx:1.16	1	admin2406
deploy4	nginx:1.17	1	admin2406
deploy5	nginx:latest	1	admin2406

```
k get deployments -n admin2406 -o json
```

```
k get deployments -n admin2406 --sort-by=.metadata.name
```

```
-o=custom-columns=DEPLOYMENT:.metadata.name,CONTAINER_IMAGE:.spec.template.spec.containers[ ].image,READY_REPLICAS:.status.readyReplicas,NAMESPACE:.metadata.namespace
```

Q&A - 3



1 2 3 4 5 6 7

Weight: 8

A kubeconfig file called `admin.kubeconfig` has been created in `/root/CKA`. There is something wrong with the configuration. Troubleshoot and fix it.

- Fix `/root/CKA/admin.kubeconfig`

Q&A - 3

```
controlplane ~ → k get po --kubeconfig=root/CKA/admin.kubeconfig
error: stat root/CKA/admin.kubeconfig: no such file or directory
```

```
controlplane ~ ✗ k get po --kubeconfig=/root/CKA/admin.kubeconfig
```

```
E1203 09:45:16.592274 14039 memcache.go:238] couldn't get current server API group list: Get "https://controlplane:4380/api?timeout=32s": dial tcp 192.18.237.9:4380: connect: connection refused
E1203 09:45:16.592601 14039 memcache.go:238] couldn't get current server API group list: Get "https://controlplane:4380/api?timeout=32s": dial tcp 192.18.237.9:4380: connect: connection refused
E1203 09:45:16.594203 14039 memcache.go:238] couldn't get current server API group list: Get "https://controlplane:4380/api?timeout=32s": dial tcp 192.18.237.9:4380: connect: connection refused
E1203 09:45:16.595605 14039 memcache.go:238] couldn't get current server API group list: Get "https://controlplane:4380/api?timeout=32s": dial tcp 192.18.237.9:4380: connect: connection refused
E1203 09:45:16.597056 14039 memcache.go:238] couldn't get current server API group list: Get "https://controlplane:4380/api?timeout=32s": dial tcp 192.18.237.9:4380: connect: connection refused
The connection to the server controlplane:4380 was refused - did you specify the right host or port?
```

```
controlplane ~ ✗ vi /root/CKA/admin.kubeconfig
```

```
controlplane ~ → k get po --kubeconfig=/root/CKA/admin.kubeconfig
```

NAME	READY	STATUS	RESTARTS	AGE
gold-nginx-6c5b9dd56c-66pnm	1/1	Running	0	10m

```
controlplane ~ →
```

```
vi /root/CKA/admin.kubeconfig #Edit port 6443
```

```
k get po --kubeconfig=/root/CKA/admin.kubeconfig
```

Q&A - 4



1 2 3 4 5 6 7

Weight: 12

Create a new deployment called `nginx-deploy`, with image `nginx:1.16` and `1` replica. Next upgrade the deployment to version `1.17` using `rolling update`.

- Image: `nginx:1.16`
- Task: Upgrade the version of the deployment to `1.17`

Q&A - 4

```
controlplane ~ → k describe deployments.apps nginx-deploy
Name: nginx-deploy
Namespace: default
CreationTimestamp: Sun, 03 Dec 2023 09:48:34 -0500
Labels: app=nginx-deploy
Annotations: deployment.kubernetes.io/revision: 2
Selector: app=nginx-deploy
Replicas: 1 desired | 1 updated | 1 total | 1 available | 0 unavailable
StrategyType: RollingUpdate
MinReadySeconds: 0
RollingUpdateStrategy: 25% max unavailable, 25% max surge
Pod Template:
  Labels: app=nginx-deploy
  Containers:
    nginx:
      Image: nginx:1.17
      Port: <none>
      Host Port: <none>
      Environment: <none>
      Mounts: <none>
      Volumes: <none>
  Conditions:
    Type      Status  Reason
    ----      -
    Available  True    MinimumReplicasAvailable
    Progressing True    NewReplicaSetAvailable
OldReplicaSets: <none>
NewReplicaSet: nginx-deploy-c848b6868 (1/1 replicas created)
Events:
  Type      Reason      Age   From          Message
  ----      -
  Normal    ScalingReplicaSet   39s   deployment-controller   Scaled up replica set nginx-deploy-7dc9dd795f to 1
  Normal    ScalingReplicaSet   2s    deployment-controller   Scaled up replica set nginx-deploy-c848b6868 to 1
  Normal    ScalingReplicaSet   1s    deployment-controller   Scaled down replica set nginx-deploy-7dc9dd795f to 0 from 1
```

```
kubectl create deployment nginx-deploy --image=nginx:1.16 --replicas=1 --dry-run=client -o yaml > 4.yaml
```

Q&A - 5



1 2 3 4 5 6 7



Weight: 20

A new deployment called `alpha-mysql` has been deployed in the `alpha` namespace. However, the pods are not running. Troubleshoot and fix the issue. The deployment should make use of the persistent volume `alpha-pv` to be mounted at `/var/lib/mysql` and should use the environment variable `MYSQL_ALLOW_EMPTY_PASSWORD=1` to make use of an empty root password.

Important: Do not alter the persistent volume.

- Troubleshoot and fix the issues

Q&A - 5

```
controlplane ~ → k edit pvc alpha-claim
error: persistentvolumeclaims "alpha-claim" is invalid
A copy of your changes has been stored to "/tmp/kubect1-edit-2329423866.yaml"
error: Edit cancelled, no valid changes were saved.
```

```
controlplane ~ ✗ k replace --force -f /tmp/kubect1-edit-2329423866.yaml
persistentvolumeclaim "alpha-claim" deleted
persistentvolumeclaim/alpha-claim replaced
```

```
controlplane ~ → k get pv,pvc
```

NAME	CAPACITY	ACCESS MODES	RECLAIM POLICY	STATUS	CLAIM	STORAGE
persistentvolume/alpha-pv	1Gi	RWO	Retain	Bound	alpha/alpha-claim	slow

NAME	STATUS	VOLUME	CAPACITY	ACCESS MODES	STORAGECLASS	AGE
persistentvolumeclaim/alpha-claim	Bound	alpha-pv	1Gi	RWO	slow	14s

```
controlplane ~ →
```

```
spec:
  accessModes: 1
  - ReadWriteOnce
  resources:
    requests: 2
    storage: 1Gi
  storageClassName: slow 3
  volumeMode: Filesystem
  volumeName: alpha-pv
status:
  accessModes:
  - ReadWriteOnce
  capacity:
    storage: 1Gi
```

B1. k describe po alpha-mysql-5b7b8988c4-f2jcm -n alpha

#Thấy error "0/2 nodes are available: persistentvolumeclaim "mysql-alpha-pvc" not found. preemption: 0/2 nodes are available: 2 No preemption victims found for incoming pod."

B3. k edit deployments.apps alpha-mysql -n alpha#Edit deployment để lấy đúng pvc name "alpha-claim"

B2. k get pv,pvc Cần check các thứ matching pvc như "accessMode, storage, storageClassName"

B3. Verify kq k get pv,pvc

Q&A - 6



1 2 3 4 5 6 7

Weight: 10

Take the backup of ETCD at the location `/opt/etcd-backup.db` on the `controlplane` node.

- Troubleshoot and fix the issues

Q&A - 6

```
controlplane ~ ❌ ETCDCCTL_API=3 etcdctl --endpoints=https://127.0.0.1:2379 \  
> --cacert=/etc/kubernetes/pki/etcd/ca.crt --cert=/etc/kubernetes/pki/etcd/server.crt --key=/etc/kubernetes/pki/etcd/server.key \  
> snapshot save /opt/etcd-backup.db  
Snapshot saved at /opt/etcd-backup.db  
  
controlplane ~ ➡ ls /opt/etcd-backup.db  
/opt/etcd-backup.db  
  
controlplane ~ ➡ []
```

```
ETCDCCTL_API=3 etcdctl --endpoints=https://127.0.0.1:2379 \  
--cacert=/etc/kubernetes/pki/etcd/ca.crt --cert=/etc/kubernetes/pki/etcd/server.crt  
--key=/etc/kubernetes/pki/etcd/server.key \  
snapshot save /opt/etcd-backup.db
```

Tips

1. k get po -n kube-system #Xem etcd chay static pod
2. cat /etc/kubernetes/manifests/etcd.yaml #Xem cac gia tri

```
-peer-trusted-ca-file=/etc/kubernetes/pki/etcd/ca.crt  
-cert-file=/etc/kubernetes/pki/etcd/server.crt  
-key-file=/etc/kubernetes/pki/etcd/server.key
```

Q&A - 7



1 2 3 4 5 6 7

Weight: 20

Create a pod called `secret-1401` in the `admin1401` namespace using the `busybox` image. The container within the pod should be called `secret-admin` and should sleep for `4800` seconds.

The container should mount a `read-only` secret volume called `secret-volume` at the path `/etc/secret-volume`. The secret being mounted has already been created for you and is called `dotfile-secret`.

☒ Pod created correctly?

Q&A - 7

```
spec:
  volumes:
    - name: secret-volume
      # secret volume
      secret:
        secretName: dotfile-secret
  containers:
    - command:
        - sleep
        - "4800"
      image: busybox
      name: secret-admin
      # volumes' mount path
      volumeMounts:
        - name: secret-volume
          readOnly: true
          mountPath: "/etc/secret-volume"
```

B1. k run secret-1401 --image=busybox -n admin1401 --dry-run=client --command sleep 4800 -o yaml > 7.yaml

B2. Edit: containerName, add thêm volume có type là Secret